INTRUSIONX SE

AI-Powered Deepfake Detection — Forensic Analysis Report

Report ID: 9B5A45CAD77A	**Generated:** 2026-04-10 17:09:42 UTC	**Classification:** CONFIDENTIAL

DETECTION VERDICT

DEEPFAKE

Confidence: 84.9%

Media Type: IMAGE

Risk Level	Critical
Anomaly Score	85%

FILE INFORMATION

Filename	`00CB415UQ7.jpg`
Content Type	`image/jpeg`

Uploaded Media Preview:

AI-GENERATED INSIGHTS

Analysis detected 1 high-severity anomalies with 84.9% confidence. Strong indicators of AI manipulation or synthetic generation were found.

	Category	Severity	Description

▲	Facial Inconsistency	HIGH	Face region analysis reveals texture irregularities consistent with GAN-generated or face-swapped imagery. Subtle artifacts detected around facial landmarks (eyes, mouth, jawline).
●	Model Disagreement	MEDIUM	Detection models produced conflicting results. One model flags manipulation while the other does not. This can occur with sophisticated deepfakes or borderline cases.

FORENSIC VISUALIZATIONS

ELA Heatmap — Error Level Analysis

Red regions indicate potential manipulation zones where compression error levels differ from surrounding areas.

Noise Variance Map

Bright regions indicate sensor noise inconsistencies. Uniform noise suggests synthetic origin.

METADATA ANALYSIS

Risk Score	30 / 100
AI Indicators	1 detected

Detected Indicators:

- No EXIF metadata found

DETECTION TELEMETRY

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■ Multi-Layer Analysis (2 models + ELA)
■ Face detected — analysed cropped face region for Model A
● deepfake_vs_real (ViT): FAKE (fake prob: 99.9%)
● AI-image-detector (Swin): REAL (fake prob: 15.6%)
■ ELA Score: 5.1 — low error levels
■ Models disagree — using cautious ensemble (max strategy)
● DEEPFAKE detected with 84.9% confidence
AI-generated or manipulated content indicators found
Cross-verify with metadata analysis below
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Models Used:

- deepfake_vs_real (ViT)
- AI-image-detector (Swin)

Model Probabilities:

Label	Score
Real (deepfake_vs_real)	0.09%
Fake (deepfake_vs_real)	99.91%
artificial (AI-image-detector)	15.58%
human (AI-image-detector)	84.42%
ELA Score	5.12%

System: Dual-Model Ensemble (ViT + Swin Transformer) + Wav2Vec2-XLSR + ELA + Metadata Forensics
Environment: CPU-only inference | Python 3.9 | FastAPI + Uvicorn

DISCLAIMER: This report is generated by an automated AI system and should be used as a forensic aid only. Results are probabilistic and should not be treated as definitive proof. Always cross-reference with manual analysis and domain expertise.